DAILY PRACTICE CALENDAR

Pre-Algebra Competition Math (Grade 6) • 2027–2028

34 weeks • Monday–Friday • 10 problems a day (about 30–40 minutes)

Experience first. Symbols second. Understanding always.

How to Use This Calendar

- **Mon–Thu:** 10 problems on the day's skill: **8 new + 2 review** from earlier weeks. Start with the **Try** problem as the warm-up.
- **Friday:** Weekly Quiz of 10 mixed problems (8 this week + 2 review). Copy every missed problem into the **error log** and redo it next Monday.
- **Holidays and breaks** are marked. A holiday's skill moves to the weekend; break weeks are not counted in the 34 weeks.
- This course follows Grade 5 Competition Math and builds toward **AMC 8 and MATHCOUNTS**. Answers to every **Try** problem are on the last page.

Year at a Glance

Wk	Dates	Topic
1	Aug 16–Aug 20	Problem-Solving Toolkit
2	Aug 23–Aug 27	Integers & the Number Line
3	Aug 30–Sep 3	Integer Operations
4	Sep 6–Sep 10	Exponents & Order of Operations
5	Sep 13–Sep 17	Primes & Prime Factorization
6	Sep 20–Sep 24	GCF & LCM Word Problems
7	Sep 27–Oct 1	Counting Factors, Squares & Cubes
8	Oct 4–Oct 8	Units Digits & Remainders
9	Oct 11–Oct 15	Divisibility & Digit Problems
10	Oct 18–Oct 22	Review & Mini-Contest 1
11	Oct 25–Oct 29	Fraction & Decimal Operations
12	Nov 1–Nov 5	Ratios & Proportions
13	Nov 8–Nov 12	Percents
14	Nov 15–Nov 19	Rates: Speed, Work & Unit Rates
	Nov 22–Nov 26	*Thanksgiving Break*
15	Nov 29–Dec 3	Averages & Statistics
16	Dec 6–Dec 10	Expressions & Variables
17	Dec 13–Dec 17	Solving Equations
	Dec 20–Dec 24	*Winter Break*
	Dec 27–Dec 31	*Winter Break*
18	Jan 3–Jan 7	Word Problems with Equations
19	Jan 10–Jan 14	Inequalities & Linear Patterns
20	Jan 17–Jan 21	Mid-Year Mock Contest & Review
21	Jan 24–Jan 28	Angles, Parallel Lines & Triangles
22	Jan 31–Feb 4	Area of Polygons
23	Feb 7–Feb 11	Circles
24	Feb 14–Feb 18	The Pythagorean Theorem
25	Feb 21–Feb 25	Volume & Surface Area
26	Feb 28–Mar 3	Coordinates & Transformations
27	Mar 6–Mar 10	Counting Principle & Arrangements
28	Mar 13–Mar 17	Combinations
29	Mar 20–Mar 24	Venn Diagrams & Complements
	Mar 27–Mar 31	*Spring Break*
30	Apr 3–Apr 7	Probability
31	Apr 10–Apr 14	Logic & Pigeonhole
32	Apr 17–Apr 21	Contest Strategy & AMC 8 Practice
33	Apr 24–Apr 28	Full Mock Contest & Review
34	May 1–May 5	Final Contest & Celebration

Units 1 & 6 · Unit 2 · Unit 3 · Unit 4 · Unit 5 · Wrap-Up · Break

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WEEK 1 • Aug 16–Aug 20, 2027

Unit 1: Integers & Number Sense

Problem-Solving Toolkit

Core: Pick a strategy before you compute: picture, simpler case, table, work backward.

Mon Aug 16

Draw a diagram

TRY A 12 m rope is cut into pieces 1.5 m long. How many cuts are made?

Tue Aug 17

Try a simpler case

TRY How many squares of all sizes are on a 4×4 grid?

Wed Aug 18

Make a table

TRY Pens cost \$3 and notebooks \$5. Maya spends exactly \$31 and buys at least one of each. How many notebooks could she buy?

Thu Aug 19

Work backward

TRY I subtract 8, divide by 3, then square the result and get 25. My number is positive. What was it?

Fri Aug 20

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 2 • Aug 23–Aug 27, 2027

Unit 1: Integers & Number Sense

Integers & the Number Line

Core: Negative numbers: opposites, distance and absolute value.

Mon Aug 23

Opposites & absolute value

TRY Compute $|-7| - |3-10| + |-2|$.

Tue Aug 24

Distance on a number line

TRY What is the distance between -13 and 9 on a number line?

Wed Aug 25

Ordering integers

TRY Order from least to greatest: $-4,\ 3,\ -11,\ 0,\ -|-6|$.

Thu Aug 26

Temperature & elevation

TRY At 6 a.m. it was $-8°$F. It rose $15°$ by noon, then fell $9°$ by night. Night temperature?

Fri Aug 27

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 3 • Aug 30–Sep 3, 2027

Unit 1: Integers & Number Sense

Integer Operations

Core: Sign rules make sense when you see the patterns behind them.

Mon Aug 30

Adding & subtracting

TRY Compute $-17 + 25 - (-8) - 30$.

Tue Aug 31

Multiplying & dividing

TRY Compute $(-6) \times (-4) \div (-3)$.

Wed Sep 1

Sign of a long product

TRY Is $(-1) \times (-2) \times (-3) \times \cdots \times (-9)$ positive or negative?

Thu Sep 2

Pairing sums

TRY Compute $1 - 2 + 3 - 4 + \cdots + 99 - 100$.

Fri Sep 3

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 4 • Sep 6–Sep 10, 2027

Unit 1: Integers & Number Sense

Exponents & Order of Operations

Core: Powers first, then multiply and divide, then add and subtract.

Mon Sep 6

HOLIDAY: Labor Day

No class. Do this skill's 10 problems on the weekend:

Powers & squares

TRY Compute $2^5 - 5^2$.

Tue Sep 7

Order of operations

TRY Compute $3 + 4 \times 2^3 - (10-4)^2 \div 9$.

Wed Sep 8

Negative bases

TRY Compute $(-2)^4 - 2^4 + (-1)^{2027}$.

Thu Sep 9

Exponent rules

TRY Write $2^3 \times 2^5 \div 2^4$ as a single power of 2.

Fri Sep 10

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 5 • Sep 13–Sep 17, 2027 — Unit 2: Number Theory

Primes & Prime Factorization **Core:** Every whole number greater than 1 is built from primes in exactly one way.

Mon Sep 13
Primes up to 100
TRY How many primes are between 40 and 70?

Tue Sep 14
Prime factorization
TRY Write 360 as a product of primes.

Wed Sep 15
Perfect-square factors
TRY What is the smallest positive integer n that makes $72n$ a perfect square?

Thu Sep 16
Factor clues
TRY The product of three consecutive integers is 2,730. What is the largest?

Fri Sep 17
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 6 • Sep 20–Sep 24, 2027 — Unit 2: Number Theory

GCF & LCM Word Problems **Core:** GCF splits into equal groups; LCM finds when things line up again.

Mon Sep 20
GCF from prime factors
TRY Find the GCF of 168 and 252.

Tue Sep 21
LCM from prime factors
TRY Find the LCM of 18, 24 and 40.

Wed Sep 22
Equal groups
TRY Ms. Lee shares 48 pencils and 72 erasers equally among kits, with none left over. What is the most kits she can make?

Thu Sep 23
When do they meet again?
TRY Two lights flash every 12 s and every 18 s. They flash together now. When do they next flash together?

Fri Sep 24
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 7 • Sep 27–Oct 1, 2027 — Unit 2: Number Theory

Counting Factors, Squares & Cubes **Core:** Count factors from exponents: add 1 to each exponent and multiply.

Mon Sep 27
Number of factors
TRY How many positive factors does 72 have?

Tue Sep 28
Perfect squares
TRY How many perfect squares are between 50 and 500?

Wed Sep 29
Perfect cubes
TRY How many positive perfect cubes are less than 1,000?

Thu Sep 30
Odd number of factors
TRY How many numbers from 1 to 50 have an odd number of factors?

Fri Oct 1
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 8 • Oct 4–Oct 8, 2027 — Unit 2: Number Theory

Units Digits & Remainders **Core:** Units digits and remainders repeat in cycles.

Mon Oct 4
Units-digit cycles
TRY What is the units digit of 3^{2027}?

Tue Oct 5
Remainder arithmetic
TRY What is the remainder when 47×53 is divided by 5?

Wed Oct 6
Day-of-the-week problems
TRY Aug 16, 2027 is a Monday. What day of the week is 100 days later?

Thu Oct 7
Remainder puzzles
TRY What is the smallest number greater than 1 that leaves remainder 1 when divided by 2, 3, 4 and 5?

Fri Oct 8
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 9 • Oct 11–Oct 15, 2027 — Unit 2: Number Theory

Divisibility & Digit Problems **Core:** Divisibility rules and digit sums unlock missing-digit problems.

Mon Oct 11
Rules for 3, 9 and 11
TRY What digit fills 5□38 so it is divisible by 9?

Tue Oct 12
Combining rules
TRY Find every digit that fills 42□6 so it is divisible by 12.

Wed Oct 13
Digit sums
TRY How many three-digit numbers have digits that add to 5?

Thu Oct 14
Palindromes
TRY How many 4-digit palindromes (like 3,773) are there?

Fri Oct 15
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 10 • Oct 18–Oct 22, 2027 — Unit 2: Number Theory

Review & Mini-Contest 1 **Core:** Pull Units 1–2 together and practice contest habits.

Mon Oct 18
Review: integers & exponents
TRY Compute $(-3)^3 + 3^3 - (-3)^2$.

Tue Oct 19
Review: primes & factors
TRY How many positive factors does $2^4 \times 3^2$ have?

Wed Oct 20
Review: cycles
TRY What is the units digit of 7^{100}?

Thu Oct 21
Timed set: 10 problems in 20 min
TRY Find the LCM of 14, 21 and 35.

Fri Oct 22
Mini-Contest 1
Units 1–2, timed. Afterward, start an error log of every missed problem.

WEEK 11 • Oct 25–Oct 29, 2027 — Unit 3: Rational Numbers, Ratios & Percents

Fraction & Decimal Operations **Core:** Fractions and decimals are numbers too: compute with them cleverly.

Mon Oct 25
Fraction arithmetic
TRY Compute $\frac{2}{3} + \frac{5}{6} - \frac{3}{4}$.

Tue Oct 26
Dividing mixed numbers
TRY Compute $3\frac{3}{4} \div 1\frac{1}{4}$.

Wed Oct 27
Telescoping sums
TRY Compute $\frac{1}{1\times2} + \frac{1}{2\times3} + \cdots + \frac{1}{9\times10}$.

Thu Oct 28
Repeating decimals
TRY Write $0.\overline{36}$ as a fraction in lowest terms.

Fri Oct 29
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 12 • Nov 1–Nov 5, 2027 — Unit 3: Rational Numbers, Ratios & Percents

Ratios & Proportions **Core:** Ratios compare parts; scale them up and down with a ratio table.

Mon Nov 1
Part-to-part & part-to-whole
TRY Red to blue marbles is 3 : 5, with 64 marbles in all. How many are red?

Tue Nov 2
Proportions
TRY If 4 notebooks cost \$7, how much do 18 notebooks cost?

Wed Nov 3
Combining ratios
TRY If $a : b = 2 : 3$ and $b : c = 4 : 5$, find $a : c$.

Thu Nov 4
Changing ratios
TRY A club of 36 has boys to girls 5 : 4. How many girls must join to make the ratio 1 : 1?

Fri Nov 5
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 13 • Nov 8–Nov 12, 2027 — Unit 3: Rational Numbers, Ratios & Percents

Percents **Core:** Percent means "per hundred." Change = original × rate.

Mon Nov 8
Percent of a number
TRY What is 15% of 260?

Tue Nov 9
Percent increase & decrease
TRY A price rises from $40 to $52. What is the percent increase?

Wed Nov 10
Finding the whole
TRY 18 is 24% of what number?

Thu Nov 11
HOLIDAY: Veterans Day
No class. Do this skill's 10 problems on the weekend:
Successive discounts
TRY An $80 item is 20% off, then 10% off the sale price. What is the final price?

Fri Nov 12
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 14 • Nov 15–Nov 19, 2027 — Unit 3: Rational Numbers, Ratios & Percents

Rates: Speed, Work & Unit Rates **Core:** Distance = speed × time; work rates add.

Mon Nov 15
Unit rates
TRY Which is the better buy: 12 oz for $3.00 or 20 oz for $4.60?

Tue Nov 16
Distance, speed & time
TRY A cyclist rides 45 km at 15 km/h, then 30 km at 10 km/h. Total time?

Wed Nov 17
Average speed
TRY You drive 60 miles at 30 mph and return at 60 mph. What is your average speed?

Thu Nov 18
Working together
TRY Ann paints a fence in 6 hours; Ben paints it in 3 hours. How long together?

Fri Nov 19
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Thanksgiving Break • Nov 22–Nov 26, 2027 • no class (not counted in the 34 weeks)

WEEK 15 • Nov 29–Dec 3, 2027 — Unit 3: Rational Numbers, Ratios & Percents

Averages & Statistics **Core:** Total = mean × count; median and mode describe the middle and the most.

Mon Nov 29
Mean
TRY Five quiz scores average 84. Four of them are 80, 90, 78 and 85. What is the fifth?

Tue Nov 30
Median & mode
TRY Find the median and mode of 7, 3, 9, 3, 12, 5, 3, 8.

Wed Dec 1
Changing the mean
TRY The mean of 6 numbers is 15. One is removed, and the mean becomes 14. Which number was removed?

Thu Dec 2
Data puzzles
TRY Five positive integers have mean 10, median 9 and a unique mode of 8. How large can the greatest one be?

Fri Dec 3
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 16 • Dec 6–Dec 10, 2027 — Unit 4: Algebra Foundations

Expressions & Variables **Core:** A letter stands for a number: write the rule, then use it.

Mon Dec 6
Writing expressions
TRY Pens cost p dollars each. Write the cost of 5 pens and a $3 notebook.

Tue Dec 7
Evaluating
TRY Find $3x^2 - 2x + 1$ when $x = -2$.

Wed Dec 8
Combining like terms
TRY Simplify $4a + 7 - 2a + 3a - 10$.

Thu Dec 9
Distributive property
TRY Simplify $3(2x - 5) - 2(x - 4)$.

Fri Dec 10
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 17 • Dec 13–Dec 17, 2027 — Unit 4: Algebra Foundations

Solving Equations — **Core:** Undo operations in reverse order; keep both sides balanced.

Mon Dec 13
One-step equations
TRY Solve $\frac{x}{4} = -7$.

Tue Dec 14
Two-step equations
TRY Solve $5x - 8 = 27$.

Wed Dec 15
Variables on both sides
TRY Solve $7x + 4 = 3x - 12$.

Thu Dec 16
Equations with fractions
TRY Solve $\frac{x}{3} + \frac{x}{6} = 5$.

Fri Dec 17
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Winter Break • Dec 20–Dec 24, 2027 • no class (not counted in the 34 weeks)

Winter Break • Dec 27–Dec 31, 2027 • no class (not counted in the 34 weeks)

WEEK 18 • Jan 3–Jan 7, 2028 — Unit 4: Algebra Foundations

Word Problems with Equations — **Core:** Name the unknown, write the equation, then check the answer in the story.

Mon Jan 3
Consecutive integers
TRY Three consecutive integers add to 147. What is the largest?

Tue Jan 4
Age problems
TRY Mia is 3 times as old as her brother. In 6 years she will be twice as old as he is. How old is Mia now?

Wed Jan 5
Coin problems
TRY Ravi has 20 coins, all dimes and quarters, worth $3.65. How many are quarters?

Thu Jan 6
Ticket problems
TRY Adult tickets cost $9 and child tickets $5. 40 tickets sold for $280. How many were adult tickets?

Fri Jan 7
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 19 • Jan 10–Jan 14, 2028 — Unit 4: Algebra Foundations

Inequalities & Linear Patterns — **Core:** Inequalities describe ranges; linear patterns grow by the same amount each step.

Mon Jan 10
Solving inequalities
TRY Solve $3x - 7 < 11$.

Tue Jan 11
Flipping the sign
TRY Solve $-2x + 5 \geq 13$.

Wed Jan 12
Arithmetic sequences
TRY Find the 50th term of 7, 11, 15, 19, . . .

Thu Jan 13
nth-term rules
TRY Figure n uses $3n + 2$ toothpicks. Which figure uses 152 toothpicks?

Fri Jan 14
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 20 • Jan 17–Jan 21, 2028 — Unit 4: Algebra Foundations

Mid-Year Mock Contest & Review — **Core:** Review Units 1–4 and practice pacing a full contest.

Mon Jan 17
HOLIDAY: MLK Jr. Day
No class. Do this skill's 10 problems on the weekend:
Review: integers & number theory
TRY What is the remainder when 2^{10} is divided by 7?

Tue Jan 18
Review: ratios & percents
TRY A shirt costs $36 after a 25% discount. What was the original price?

Wed Jan 19
Review: equations
TRY Solve $2(x + 3) = 5x - 9$.

Thu Jan 20
Pacing: skip & return
TRY 15-minute set. Warm-up: the mean of -4, 7, 10 and 15?

Fri Jan 21
Mid-Year Mock Contest
Full-length, Units 1–4. Then sort errors by topic in the error log.

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WEEK 21 • Jan 24–Jan 28, 2028 — Unit 5: Geometry & Measurement

Angles, Parallel Lines & Triangles

Core: Angles on a line add to $180°$; parallel lines copy angles.

Mon Jan 24
Complements & supplements
TRY An angle is 4 times its complement. Find the angle.

Tue Jan 25
Parallel lines & transversals
TRY Corresponding angles measure $(3x + 10)°$ and $(5x - 30)°$. Find x.

Wed Jan 26
Triangle angle sum
TRY The angles of a triangle are in the ratio $2:3:4$. Find the largest angle.

Thu Jan 27
Polygon angles
TRY Each interior angle of a regular polygon is $150°$. How many sides does it have?

Fri Jan 28
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 22 • Jan 31–Feb 4, 2028 — Unit 5: Geometry & Measurement

Area of Polygons

Core: Split or surround odd shapes with rectangles and triangles.

Mon Jan 31
Triangles & parallelograms
TRY A triangle has base 14 and area 63. What is its height?

Tue Feb 1
Trapezoids
TRY A trapezoid has bases 8 and 14 and height 5. Find its area.

Wed Feb 2
Composite figures
TRY A 4×3 rectangle is cut from a corner of a 10×8 rectangle. Find the area and perimeter of what is left.

Thu Feb 3
Areas on a grid
TRY Find the area of the triangle with vertices $(0, 0)$, $(8, 0)$ and $(3, 6)$.

Fri Feb 4
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 23 • Feb 7–Feb 11, 2028 — Unit 5: Geometry & Measurement

Circles

Core: $C = \pi d$ and $A = \pi r^2$; leave answers in terms of π.

Mon Feb 7
Circumference
TRY A circle has diameter 14. Find its circumference.

Tue Feb 8
Area
TRY A circle has circumference 10π. Find its area.

Wed Feb 9
Shaded regions
TRY The largest possible circle sits inside a square of side 10. Find the area inside the square but outside the circle.

Thu Feb 10
Sectors
TRY A sector has radius 6 and central angle $60°$. Find its area.

Fri Feb 11
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 24 • Feb 14–Feb 18, 2028 — Unit 5: Geometry & Measurement

The Pythagorean Theorem

Core: $a^2 + b^2 = c^2$ in a right triangle; learn the common triples.

Mon Feb 14
Finding the hypotenuse
TRY A right triangle has legs 9 and 12. Find the hypotenuse.

Tue Feb 15
Finding a leg
TRY The hypotenuse is 26 and one leg is 10. Find the other leg.

Wed Feb 16
Pythagorean triples
TRY Is a triangle with sides 20, 21 and 29 a right triangle?

Thu Feb 17
Distance on a grid
TRY Find the distance between $(-2, 3)$ and $(4, 11)$.

Fri Feb 18
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 25 • Feb 21–Feb 25, 2028 — Unit 5: Geometry & Measurement

Volume & Surface Area **Core:** Volume fills a solid; surface area wraps it.

Mon Feb 21

HOLIDAY: Presidents' Day

No class. Do this skill's 10 problems on the weekend:

Volume of prisms

TRY Find the volume of a $6 \times 4 \times 5$ box.

Tue Feb 22

Surface area of prisms

TRY Find the surface area of a $6 \times 4 \times 5$ box.

Wed Feb 23

Cylinders

TRY A cylinder has radius 3 and height 10. Find its volume.

Thu Feb 24

Scaling

TRY Every edge of a cube is doubled. Its volume is multiplied by what?

Fri Feb 25

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 26 • Feb 28–Mar 3, 2028 — Unit 5: Geometry & Measurement

Coordinates & Transformations **Core:** Plot it first, then slide, flip or turn.

Mon Feb 28

Midpoints

TRY Find the midpoint of $(-3, 5)$ and $(7, -1)$.

Tue Feb 29

Reflections

TRY Reflect $(4, -3)$ over the y-axis.

Wed Mar 1

Rotations

TRY Rotate $(2, 5)$ by $90°$ counterclockwise about the origin.

Thu Mar 2

Slope as a rate

TRY Find the slope of the line through $(1, 2)$ and $(5, 14)$.

Fri Mar 3

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 27 • Mar 6–Mar 10, 2028 — Unit 6: Counting, Probability & Logic

Counting Principle & Arrangements **Core:** Choices in stages multiply; in an arrangement, order matters.

Mon Mar 6

Multiplication principle

TRY You have 3 shirts, 4 pants and 2 pairs of shoes. How many outfits?

Tue Mar 7

Arrangements

TRY How many ways can the letters of MATH be arranged?

Wed Mar 8

Restrictions

TRY How many three-digit numbers have all different digits?

Thu Mar 9

Repeated letters

TRY How many ways can the letters of LEVEL be arranged?

Fri Mar 10

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 28 • Mar 13–Mar 17, 2028 — Unit 6: Counting, Probability & Logic

Combinations **Core:** When order does not matter, divide out the repeats.

Mon Mar 13

Choosing a group

TRY How many ways can you choose 3 students from 7?

Tue Mar 14

Handshakes & diagonals

TRY How many diagonals does an octagon have?

Wed Mar 15

Paths on a grid

TRY Moving only right or up, how many shortest paths cross a 4×3 grid of blocks from corner to corner?

Thu Mar 16

Committees

TRY Choose a committee of 2 boys and 2 girls from 5 boys and 4 girls. How many ways?

Fri Mar 17

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 29 • Mar 20–Mar 24, 2028 — Unit 6: Counting, Probability & Logic

Venn Diagrams & Complements — **Core:** Count overlaps once; sometimes count what you don't want and subtract.

Mon Mar 20
Two-set Venn diagrams
TRY In a class of 30, 18 play soccer, 15 play basketball and 5 play neither. How many play both?

Tue Mar 21
Inclusion–exclusion
TRY How many integers from 1 to 100 are divisible by 3 or 5?

Wed Mar 22
Complementary counting
TRY How many three-digit numbers have at least one digit 7?

Thu Mar 23
Three conditions
TRY How many integers from 1 to 60 are divisible by none of 2, 3 and 5?

Fri Mar 24
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Spring Break • Mar 27–Mar 31, 2028 • no class (not counted in the 34 weeks)

WEEK 30 • Apr 3–Apr 7, 2028 — Unit 6: Counting, Probability & Logic

Probability — **Core:** Probability = favorable outcomes ÷ all equally likely outcomes.

Mon Apr 3
Single events
TRY A number from 1 to 30 is picked at random. What is the probability it is prime?

Tue Apr 4
Two dice
TRY Two dice are rolled. What is the probability the sum is 8?

Wed Apr 5
Complements
TRY Four coins are flipped. What is the probability of at least one head?

Thu Apr 6
Draws without replacement
TRY A bag has 3 red and 5 blue marbles. Two are drawn without replacement. P(both red)?

Fri Apr 7
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 31 • Apr 10–Apr 14, 2028 — Unit 6: Counting, Probability & Logic

Logic & Pigeonhole — **Core:** Reason from clues; guarantee results with pigeonholes.

Mon Apr 10
Truth-tellers & liars
TRY Each person always tells the truth or always lies. A says, "B is a liar." B says, "A and I are both liars." Who tells the truth?

Tue Apr 11
Pigeonhole principle
TRY A dark drawer has 6 red, 8 blue and 10 black socks. How many must you take to be sure of a matching pair?

Wed Apr 12
Worst case
TRY Same drawer: how many must you take to be sure of 2 blue socks?

Thu Apr 13
Logic grids
TRY Amy, Ben and Cal own a cat, a dog and a fish. Amy's pet has no fur. Ben does not own the dog. Who owns which pet?

Fri Apr 14
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 32 • Apr 17–Apr 21, 2028 — Contest Season Wrap-Up

Contest Strategy & AMC 8 Practice — **Core:** Read every choice, estimate, eliminate, and test answers when stuck.

Mon Apr 17
Estimate & eliminate
TRY Which is closest to $0.48 \times 3{,}012 \div 19.8$: 7, 70, 700 or 7,000?

Tue Apr 18
Test the choices
TRY A number plus its square is 156. Which is it: 10, 11, 12 or 13?

Wed Apr 19
AMC 8 style (Problems 1–10)
TRY Compute $1+2-3+4+5-6+\cdots+22+23-24$.

Thu Apr 20
Timed set: 10 in 20 min
TRY How many positive integers less than 200 are multiples of 6 but not of 9?

Fri Apr 21
Weekly Quiz
10 mixed contest problems, timed (25 min).

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WEEK 33 • Apr 24–Apr 28, 2028 — Contest Season Wrap-Up

Full Mock Contest & Review

Core: Simulate contest day and learn from every error.

Mon Apr 24

Review: number theory

TRY What is the units digit of $13^{13} + 7^7$?

Tue Apr 25

Review: geometry

TRY A right triangle has legs 5 and 12. Find its perimeter.

Wed Apr 26

Review: algebra

TRY Solve $3(x - 2) = x + 10$.

Thu Apr 27

Full Mock Contest

Full-length mixed contest under real time limits.

Fri Apr 28

Error-log review

Redo every missed problem from the mock; write the key idea for each.

WEEK 34 • May 1–May 5, 2028 — Contest Season Wrap-Up

Final Contest & Celebration

Core: Show what you have learned and set next year's goals.

Mon May 1

Light review: favorite strategies

TRY Find $2 + 4 + 6 + \cdots + 40$.

Tue May 2

Mixed warm-up set

TRY Find the LCM of 12 and 18 minus their GCF.

Wed May 3

Final Contest

Full-length contest covering the whole year.

Thu May 4

Contest review

Go over solutions together; students present one problem each.

Fri May 5

Celebration & reflection

Certificates, progress review, and goals for next year.

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ANSWER KEY

Answers to the Daily "Try" Problems

Week 1
Mon: 7 cuts
Tue: 30
Wed: 2 or 5
Thu: 23

Week 2
Mon: 2
Tue: 22
Wed: $-11, -|-6|, -4, 0, 3$
Thu: $-2°$F

Week 3
Mon: -14
Tue: -8
Wed: Negative (9 negative factors)
Thu: -50

Week 4
Mon: 7
Tue: 31
Wed: -1
Thu: $2^4 = 16$

Week 5
Mon: 7
Tue: $2^3 \times 3^2 \times 5$
Wed: 2
Thu: 15

Week 6
Mon: 84
Tue: 360
Wed: 24
Thu: In 36 s

Week 7
Mon: 12
Tue: 15
Wed: 9
Thu: 7 (the perfect squares)

Week 8
Mon: 7
Tue: 1
Wed: Wednesday
Thu: 61

Week 9
Mon: 2
Tue: 3 or 9
Wed: 15
Thu: 90

Week 10
Mon: -9
Tue: 15
Wed: 1
Thu: 210

Week 11
Mon: $\frac{3}{4}$
Tue: 3
Wed: $\frac{9}{10}$
Thu: $\frac{4}{11}$

Week 12
Mon: 24
Tue: \$31.50
Wed: 8 : 15
Thu: 4

Week 13
Mon: 39
Tue: 30%
Wed: 75
Thu: \$57.60

Week 14
Mon: 20 oz (\$0.23/oz)
Tue: 6 hours
Wed: 40 mph
Thu: 2 hours

Week 15
Mon: 87
Tue: Median 6, mode 3
Wed: 20
Thu: 15

Week 16
Mon: $5p + 3$
Tue: 17
Wed: $5a - 3$
Thu: $4x - 7$

Week 17
Mon: $x = -28$
Tue: $x = 7$
Wed: $x = -4$
Thu: $x = 10$

Week 18
Mon: 50
Tue: 18
Wed: 11
Thu: 20

Week 19
Mon: $x < 6$
Tue: $x \le -4$
Wed: 203
Thu: Figure 50

Week 20
Mon: 2
Tue: \$48
Wed: $x = 5$
Thu: 7

Week 21
Mon: $72°$
Tue: $x = 20$
Wed: $80°$
Thu: 12

Week 22
Mon: 9
Tue: 55
Wed: Area 68, perimeter 36
Thu: 24

Week 23
Mon: 14π
Tue: 25π
Wed: $100 - 25\pi$
Thu: 6π

Week 24
Mon: 15
Tue: 24
Wed: Yes: $400 + 441 = 841$
Thu: 10

Week 25
Mon: 120
Tue: 148
Wed: 90π
Thu: 8

Week 26
Mon: $(2, 2)$
Tue: $(-4, -3)$
Wed: $(-5, 2)$
Thu: 3

Week 27
Mon: 24
Tue: 24
Wed: 648
Thu: 30

Week 28
Mon: 35
Tue: 20
Wed: 35
Thu: 60

Week 29
Mon: 8
Tue: 47
Wed: 252
Thu: 16

Week 30
Mon: $\frac{1}{3}$
Tue: $\frac{5}{36}$
Wed: $\frac{15}{16}$
Thu: $\frac{3}{28}$

Week 31
Mon: A tells the truth; B lies
Tue: 4
Wed: 18
Thu: Amy fish, Ben cat, Cal dog

Week 32
Mon: 70
Tue: 12
Wed: 84
Thu: 22

Week 33
Mon: 6
Tue: 30
Wed: $x = 8$

Week 34
Mon: 420
Tue: 30

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